TEC-C ARCHITECTURE & EXECUTIVE PLAIN-ENGLISH GUIDE

How Tec-C Works: Complete System & Flow

Autonomous AI Capital Management with User-Tuned Blockchain Leashes

> **The Core Metaphor:** Imagine hiring a top-tier Wall Street quantitative trader to manage your crypto capital, but instead of giving them blank checks, you hand them a corporate card on an **intelligent mathematical leash**. **You choose your own daily spending limit** (e.g. $250, $1,000, or $5,000/day). Within that radius and on vetted blue-chip protocols (Aave, Uniswap), the AI trades autonomously 24/7. If it spots a breakout opportunity exceeding your daily allowance, it cannot spend a single penny without your **FaceID / TouchID biometric approval**.

01 The Three Foundation Pillars of Tec-C

1. THE AI BRAIN

The Cognitive Driver. Interprets your plain English instructions, reads live market sentiment (Fear & Greed, liquidation risk, news), scans yield pools, and captures stablecoin arbitrage.

2. THE SMART VAULT ("THE TETHER")

The Blockchain Leash. An ERC-4337 smart wallet enforcing your chosen daily limits, protocol whitelists, and risk tiers directly on-chain before any execution can occur.

3. MOBILE VIRTUAL SESSION

The Master Remote Key. Runs locally on your laptop, but projects a private encrypted session to your smartphone so you can monitor, steer, and approve trades anywhere in the world.

02 The Big Problem Tec-C Solves

Before Tec-C, crypto investors had to choose between two unacceptable extremes:

Feature / Risk	Traditional Trading Bots	Standard Hardware Wallet	Tec-C Autonomous Vault
Autonomous 24/7 Trading	✓ Full autonomy	✕ Zero (Manual click for everything)	✓ Full autonomy inside your limit
Catastrophic Hack Risk	✕ High (API keys can drain 100%)	✓ Low (Keys kept offline)	✓ Zero (Capped by on-chain leash)
Daily Allowance Tuning	✕ Unbounded	✕ Not applicable	✓ You choose & AI adapts to accuracy
Seed Phrase Vulnerability	✕ Vulnerable to leaks & phishing	✕ 12 words written on paper	✓ None (Hardware FaceID / Passkeys)
Stablecoin Crash Protection	✕ Blind to depegs	✕ Too slow (You must be awake)	✓ PegGuard auto-swaps before loss

03 Dynamic Elasticity: The Accuracy Feedback Loop

🔄 Personalized Limit: The AI Earns Autonomy with Accuracy — User-Configured Feature

When setting up Tec-C, you select your initial daily spending limit (e.g. $1,000/day). Then you are asked: "*Do you want your limit to be personalized and adaptively influenced by AI accuracy?*"

✓ High AI Accuracy (Rewards):
When the AI executes profitable trades and you ratify them in the Morning Review, the vault awards a step-up (up to **+10% per day max**) towards your configured hard ceiling.

✕ Mistakes & Disapprovals (Penalties):
If the AI makes a bad trade, flagging it immediately contracts the daily limit by **-20%**. Adding a penalty guidance note contracts it by **-10%**. Trust is lost 7x faster than it is gained!

04 Step-by-Step Flow: How a User Command Becomes a Trade

Here is the exact journey of what happens behind the scenes from the moment you speak or type a command until profits hit your vault:

1 **User Command in Plain English** `Input`

You type or speak naturally into the app: "*Find the best yield for my USDC on blue-chip pools, but avoid Curve.*" You don't need to know complex DeFi routing or smart contract methods.

2 **Security Firewall & Privacy Sieve** `Safety`

Before your words reach any AI model, a local security firewall verifies that the prompt has no malicious prompt-injection attacks. It converts your request into a strict mathematical **Mandate**: `Asset: USDC | Action: YIELD | Avoid: Curve`.

3 **Sentinel Market Weather Radar** `Intelligence`

The Brain scans live market indicators: **CMC Fear & Greed Index** (market mood), **Coinglass** (whale leverage & liquidation dangers), and **Real-time News** (exploits/breaking events). If a flash crash or market panic is underway, it automatically tightens risk limits.

4 **Opportunity Search & PegGuard Scanning** `Discovery`

The Brain scans whitelisted protocols (Aave, Compound, Uniswap, Spark, Morpho) for top yields. Concurrently, **PegGuard** inspects 25+ stablecoins across DEXs for depeg risks and price-difference arbitrage opportunities.

5 **Hard Risk Filters & Asymmetric Scoring** `Risk Math`

Every candidate opportunity must pass strict math tests: pool depth, smart contract audit rating, and depeg probability. The math model uses *loss-averse scoring* (downside risk is penalized 2.5x more than upside gain). Surviving candidates are ranked as **Good, Better, and Best**.

6 **Bounded AI Decision & Plain-English Explanation** `Synthesis`

The LLM evaluates the pre-vetted options. **The Golden Safety Rule:** The LLM can choose among candidates or reduce trade size, but it can *never* add unapproved venues or spend more money. It generates a clear rationale: "*Selected Aave v3 USDC at 9.2% APY with $45M reserve safety. Avoided Curve per your instructions.*"

7 **The Tether Gatekeeper & On-Chain Execution** `Execution`

The trade is dispatched to your Smart Vault:

• **Within User's Daily Limit:** Executes automatically on-chain. Zero friction.

• **Breakout Trade (Exceeds Limit):** Pauses and sends a prompt to your mobile phone. You scan FaceID, opening a 5-minute window that completes the trade.

05 PegGuard: Stablecoin Radar & Arbitrage

How PegGuard Protects & Grows Your Balance:

Stablecoins are supposed to stay at exactly $1.000. But during turbulence, they fluctuate:

- **Crash Defense:** If a coin you hold slips below $0.985 (like UST in 2022), PegGuard immediately detects the anomaly and swaps your balance into safe blue-chip collateral before you lose money.
- **Risk-Free Arbitrage:** If DAI drops temporarily to $0.993 on one exchange while trading at $1.001 on another, PegGuard captures the price gap automatically, earning free yield while keeping your money in dollars.

06 The User Experience (UX) Flow: What You Actually See & Do

Tec-C was designed so that anyone can operate sophisticated institutional trading strategies without touching complex blockchain tooling. Here is your day-to-day experience:

PHASE 1
60 Seconds

1. Custom Onboarding & Personalization Opt-In

You open the app and tap FaceID/TouchID. No seed phrases to write down. Next, you set your own customized daily limit and decide whether you want the AI to earn or lose allowance based on its accuracy:

> **[Onboarding Setup] Configure Your Personal Leash** — **Custom Tier**
> Initial Daily Limit: **$1,000 / day** (User-Chosen) • Hard Ceiling: **$3,000 / day**
> Adaptive Personalization: **[x] ENABLED** *(AI limit expands with high accuracy, shrinks on errors)*

PHASE 2
Interactive

2. Conversational Command Center (The Interactive Dashboard)

You interact with your portfolio like a private hedge fund chat. You type or speak commands. The AI responds with simple, clean **Decision Cards** comparing vetted opportunities:

> 🤖 **Tec-C Brain Verdict: Found 3 Candidates** — **Market: Neutral (62/100)**
> • **[BEST] Aave v3 USDC:** 9.4% APY | Safety Score: 96/100 | *"Meets your avoid:Curve constraint."*
> • **[BETTER] Spark DAI:** 8.2% APY | Safety Score: 92/100 | Deep reserve liquidity.
> **Action:** Autonomous execution dispatched ($400 under daily limit).

PHASE 3
Morning Routine

3. Morning Review: Teach, Reward, or Penalize Your AI

Every morning, the app gives you a 15-second summary of what the AI traded while you slept. This is where you directly influence its future limits:

> 🌞 **Morning Review: 2 Autonomous Executions** — **Status: Pending Review**
> • **Trade #1:** Lent $400 on Aave v3 (+ $1.20 yield earned) ➔ **[👍 Ratify / Like]** *(Recharges radius & builds trust)*
> • **Trade #2:** Rebalanced pool ➔ **[👎 Flag (-20% Penalty)]** or **[📝 Add Note (-10%)]**

PHASE 4
Anywhere

4. Mobile Virtual Session & 2-Second Breakout Approval

Click *"Create Virtual Session"* on your laptop, scan the QR code with your phone camera, and authenticate with mobile FaceID. The session stays live all day. If a high-value trade exceeds your limit:

> 🔔 **PUSH ALERT: High-Profit Breakout Opportunity** — **Just Now**
> *"AI Brain spotted a $4,500 USDC/DAI arbitrage for +$125 net profit (exceeds your $1,000 daily limit)."*
> **[Confirm with FaceID]** ➔ **5-Minute Window Opens & Trade Executes!**

PHASE 5
Instant

5. One-Tap Panic Button (Total Peace of Mind)

Both your desktop dashboard and mobile phone screen feature a prominent red **Emergency Freeze** button. Tapping it instantly revokes the AI session key's execution permissions on-chain.

07 Real-World Scenarios: Tec-C in Action

Here is how Tec-C behaves in everyday life under different market conditions:

Scenario A: The Silent Daily Yield Earner

100% Autonomous

Situation: You set an initial daily limit of $1,000 for your $10,000 portfolio.

AI Discovery: The AI Brain spots that Compound v3 is offering a flash 11.2% lending yield on USDC.

What Happens: The AI allocates $800 into the pool. Because $800 is under your $1,000 daily leash, the smart contract executes the deposit silently on-chain.

User Experience: You wake up in the morning, check your dashboard, and see that you earned extra yield while you were asleep—without having to approve a thing.

Scenario B: The High-Value Breakout Opportunity

Biometric FaceID Required

Situation: Your daily limit is $1,000. The AI Brain spots a massive arbitrage opportunity that requires a $4,500 swap to capture a $120 profit in 30 seconds.

What Happens: The smart contract detects that $4,500 exceeds your $1,000 leash. It blocks autonomous execution and sends an urgent push alert to your phone.

User Action: Your phone buzzes: "*High-Profit Arbitrage Opportunity: $4,500 swap to gain $120. Tap to approve.*"

Result: You glance at your screen, verify the profit, scan FaceID, and the 5-minute biometric window opens. The trade finishes instantly.

Scenario C: Dynamic Elasticity in Action (AI Gains & Loses Trust)

Adaptive Learning

Starting Point: You configure a $1,000 daily limit with Dynamic Elasticity enabled (Hard Ceiling: $2,500).

Week 1 (High Accuracy): The AI executes 5 successful yield trades. You hit "Ratify" in Morning Review. The vault increases your daily allowance by +10%/day, reaching $1,460.

Week 2 (User Correction): The AI enters a pool you dislike. In Morning Review, you flag it for revert. The smart contract immediately penalizes the AI by **-20%**, shrinking the limit back down to $1,168.

Outcome: The AI learns your exact risk appetite, and its spending power reflects your ongoing satisfaction.

Scenario D: PegGuard Defense in a Market Panic

Crisis Circuit-Breaker

Situation: You hold 6,000 units of a synthetic stablecoin. Breaking news hits that the protocol's backing reserves are under stress.

Sentinel Alert: The market radar detects social panic and rising liquidation cascades; the stablecoin price slips to $0.982.

PegGuard Action: Within seconds, PegGuard activates its automated defense protocol and swaps the 6,000 synthetic coins into native USDC at $0.981.

Result: An hour later, the troubled stablecoin crashes to $0.40. While other investors lost 60% of their money, your vault protected 98.1% of your capital.

Scenario E: The Lost Phone Emergency

Zero Seed Phrase Risk

Situation: You accidentally drop your iPhone in the ocean. How do you recover your money without a 12-word seed phrase?

Multi-Device Passkey: Because you linked both your iPhone and your MacBook TouchID during setup, you can immediately open your MacBook, log into your vault with TouchID, and revoke the lost iPhone's access key.

Emergency Guardian: Even if both devices were destroyed in a house fire, your pre-configured Recovery Guardian (a trusted hardware key or family multisig) can sign an authorization to attach a brand-new device. Your funds are never lost.